

Complete-Split Extremizers for Fractional Clique Parameters on Chordal Graphs

Juan Pablo Traverso Gianini

Independent researcher, Santiago, Chile

jtraverso@gmail.com

ORCID: [0009-0003-6068-4096](https://orcid.org/0009-0003-6068-4096)

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Closed author version; the proof is analytic modulo cited mathematical theorems; extensive multi-model AI-assisted adversarial checking completed; not independently peer-reviewed or journal-refereed; forward-citation and novelty tracking remain pending.

Abstract

Balogh, He, Krueger, Nguyen, and Wigal developed a general framework for fractional clique covers and clique decompositions based on Zykov symmetrization. We prove a constrained version for chordal graphs. If two nonadjacent maximal clone classes contain simplicial vertices, symmetrizing either class to the other preserves chordality; their fractional symmetrization lemma guarantees a nondecreasing direction, and every step merges clone classes.

The terminal chordal graphs are precisely the complete split graphs

$$S_{p,q} := K_p \vee \overline{K}_q.$$

Hence every isomorphism-invariant clone-symmetrization-monotone parameter attains its maximum over n -vertex chordal graphs on one of the $n+1$ graphs $S_{p,n-p}$. The same maximum is therefore attained inside the threshold graphs. In particular, this applies to all fractional clique-cover and clique-decomposition parameters $CC_t^*(G, \mathbf{c})$ and $CD_t^*(G, \mathbf{c})$ in the Balogh–He–Krueger–Nguyen–Wigal framework.

For

$$\Phi_r^*(G) = |E(G)| - \left(\binom{r}{2} - 1 \right) \nu_r^*(G),$$

the reduction leaves a two-variable orbit linear program on $S_{p,q}$. Solving it gives an exact finite formula for every $r \geq 3$: the optimum is the larger of an explicit K_r -free candidate and the nearest admissible integer to the vertex of the saturated quadratic. For triangles this specializes to

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} (|E(G)| - 2\nu_3^*(G)) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

For every fixed r , the exact formula implies

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \Phi_r^*(G) = \frac{r-1}{4r} n^2 + O_r(n).$$

As consequences, the maximum number of parts in a K_r -and-edge partition of an n -vertex chordal graph is

$$\left(\frac{r-1}{4r} + o(1) \right) n^2,$$

and a complete-split extremizer can be constructed in exact polynomial time for fixed r . The structural reduction and finite fractional optimization are exact; the $o(n^2)$ loss enters only in the final fractional-to-integral packing theorem.

Keywords: chordal graph; complete split graph; threshold graph; clique cover; clique decomposition; fractional clique packing; Zykov symmetrization; clone classes; simplicial vertex.

1. Introduction

A split graph is a graph whose vertex set can be partitioned into a clique and an independent set. A **complete split graph** is a graph of the form

$$S_{p,q} := K_p \vee \overline{K}_q, \tag{1.1}$$

where every clique vertex is adjacent to every independent vertex.

Erdős, Ordman, and Zalcstein asked for the asymptotic maximum of the edge-clique partition number $\text{cp}(G)$ over n -vertex chordal graphs. The complete split constructions in the split-graph literature already show that the leading constant is at least $1/6$.

The starting point of this paper is the fractional clone-symmetrization framework of Balogh, He, Krueger, Nguyen, and Wigal. Their unrestricted process produces complete multipartite extremizers. We restrict the operation to maximal clone classes containing simplicial vertices. This keeps the graph chordal and leads to a different canonical terminal family: the complete split graphs.

The paper separates into three layers.

1. **Structure.** Constrained symmetrization reduces every relevant chordal extremal problem to $S_{p,n-p}$.
2. **Analysis.** On $S_{p,q}$, the fixed- K_r packing problem becomes a two-variable orbit linear program.
3. **Applications.** Solving that program gives the exact finite K_r theorem, its triangle specialization, the integral K_r -and-edge asymptotics, and the constructive fixed- r reduction.

Once the complete-split reduction has been established, every quantitative result in the paper follows from the two-orbit program. This is a worst-case extremal reduction, not a typical-graph statement; it is therefore distinct from the theorem that almost all labelled chordal graphs are split.

1.1 Structural result

Let $\mathcal{C}_n$ be the family of chordal graphs on a fixed n -vertex set, and let

$$\mathcal{Q}_n = \{S_{p,n-p} : 0 \leq p \leq n\}.$$

A graph parameter F is **clone-symmetrization monotone** if, whenever U, V are disjoint nonadjacent clone classes,

$$F(G) \leq \max\{F(G(U \rightarrow V)), F(G(V \rightarrow U))\}. \quad (1.2)$$

Theorem 1.1 — Complete-split extremizer principle

Let F be an isomorphism-invariant clone-symmetrization-monotone graph parameter. For every chordal graph G , there is a complete split graph $S_{p,n-p}$ on the same vertex set such that

$$F(S_{p,n-p}) \geq F(G). \quad (1.3)$$

Consequently,

$$\boxed{\max_{G \in \mathcal{C}_n} F(G) = \max_{0 \leq p \leq n} F(S_{p,n-p})}. \quad (1.4)$$

1.2 Universal fractional consequence

For every integer $t \geq 1$ and every nonnegative clique-cost sequence $\mathbf{c}$, let

$$CC_t^*(G, \mathbf{c}) \quad \text{and} \quad CD_t^*(G, \mathbf{c})$$

denote the fractional t -clique cover and fractional t -clique decomposition parameters of Balogh et al.

Corollary 1.2

For every $t \geq 1$, every nonnegative cost sequence $\mathbf{c}$, and every n ,

$$\max_{G \in \mathcal{C}_n} CC_t^*(G, \mathbf{c}) = \max_{0 \leq p \leq n} CC_t^*(S_{p, n-p}, \mathbf{c}), \quad (1.5)$$

and

$$\max_{G \in \mathcal{C}_n} CD_t^*(G, \mathbf{c}) = \max_{0 \leq p \leq n} CD_t^*(S_{p, n-p}, \mathbf{c}). \quad (1.6)$$

Thus an extremal optimization over all chordal graphs reduces to at most $n + 1$ complete split graphs.

Since every complete split graph is a threshold graph and every threshold graph is chordal, Theorem 1.1 also gives a threshold-graph sandwich; see Corollary 5.1.

1.3 Quantitative fixed-motif results

For $r \geq 3$, let $\nu_r^*(G)$ be the fractional packing number of edge-capacitated copies of K_r , and define

$$\Phi_r^*(G) = |E(G)| - \left(\binom{r}{2} - 1 \right) \nu_r^*(G). \quad (1.7)$$

Corollary 1.3

For every $r \geq 3$ and every n ,

$$\max_{G \in \mathcal{C}_n} \Phi_r^*(G) = \max_{0 \leq p \leq n} \Phi_r^*(S_{p, n-p}). \quad (1.8)$$

The right-hand side is evaluated explicitly in Section 6.

Theorem 1.4 — Exact triangle maximum

For every $n \geq 1$,

$$\max_{G \in \mathcal{C}_n} (|E(G)| - 2\nu_3^*(G)) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor. \quad (1.9)$$

Theorem 1.5 — Exact finite K_r maximum

Fix $r \geq 3$ and $n \geq 1$. Put

$$\ell = \min\{n, r-2\}, \quad L_{r,n} = \frac{\ell(2n - \ell - 1)}{2}. \quad (1.10)$$

If $n \geq r$, define

$$p_0 = \frac{(r-1)n+1}{2r}, \quad d_{r,n} = \min_{\substack{p \in \mathbb{Z} \\ p \geq r-1}} |p - p_0|, \quad (1.11)$$

and

$$A_{r,n}^\# = \frac{((r-1)n+1)^2}{4r(r-1)} - \frac{r}{r-1} d_{r,n}^2. \quad (1.12)$$

Then

$$\max_{G \in \mathcal{C}_n} \Phi_r^*(G) = \begin{cases} L_{r,n}, & n < r, \\ \max\{L_{r,n}, A_{r,n}^\#\}, & n \geq r. \end{cases} \quad (1.13)$$

In particular, for every fixed $r \geq 3$,

$$\max_{G \in \mathcal{C}_n} \Phi_r^*(G) = \frac{r-1}{4r} n^2 + O_r(n). \quad (1.14)$$

1.4 Integral application

Let $\pi_r(G)$ be the minimum number of parts in a partition of $E(G)$ into copies of K_r and single edges. Equivalently,

$$\pi_r(G) = |E(G)| - \left(\binom{r}{2} - 1 \right) \nu_r(G). \quad (1.15)$$

Corollary 1.6

For every fixed $r \geq 3$,

$$\max_{G \in \mathcal{C}_n} \pi_r(G) = \left(\frac{r-1}{4r} + o(1) \right) n^2. \quad (1.16)$$

For $r = 3$, this gives the upper bound needed for the chordal clique-partition problem:

$$\text{cp}(G) \leq \pi_3(G) \leq \left(\frac{1}{6} + o(1) \right) n^2. \quad (1.17)$$

The known complete split construction gives the matching lower bound for cp.

1.5 Constructive consequence

Corollary 1.7

For every fixed $r \geq 3$, there is an exact polynomial-time algorithm that, given an n -vertex chordal graph G , constructs an n -vertex complete split graph $S_{p,n-p}$ satisfying

$$\Phi_r^*(S_{p,n-p}) \geq \Phi_r^*(G). \quad (1.18)$$

The degree of the polynomial depends on r . No polynomial-time claim is made when r is part of the input or for arbitrary unrestricted cost sequences in Corollary 1.2.

1.6 Organization

Section 2 recalls the fractional symmetrization framework and identifies Φ_r^* as a clique-decomposition cost. Sections 3–5 prove the constrained chordal reduction and the complete-split terminal characterization. Section 6 solves the complete-split orbit program. Sections 7–9 derive the exact triangle, exact finite fixed- r , and integral consequences. Sections 10–12 record the fractional limitation, computational audit, attribution, limitations, and future directions.

2. Fractional framework and prior symmetrization

Fix $t \geq 1$ and a nonnegative cost sequence

$$\mathbf{c} = (c_i)_{i \geq t}.$$

A fractional t -clique cover assigns a nonnegative weight w_Q to every clique Q of order at least t , so that every copy T of K_t satisfies

$$\sum_{Q \supseteq T} w_Q \geq 1.$$

Its cost is

$$\sum_Q c_{|Q|} w_Q.$$

The minimum cost is $CC_t^*(G, \mathbf{c})$.

A fractional t -clique decomposition replaces the cover inequalities by equalities:

$$\sum_{Q \supseteq T} w_Q = 1.$$

Its minimum cost is $CD_t^*(G, \mathbf{c})$.

Two vertices are **clones** when they have the same open neighborhood. Clone classes are independent sets.

Let U, V be disjoint clone classes with no edges between them. The graph $G(U \rightarrow V)$ is obtained by deleting every edge incident with U and then making every vertex of U a false twin of the vertices of V .

The following is the symmetrization lemma of Balogh, He, Krueger, Nguyen, and Wigal.

Lemma 2.1 — Fractional clone symmetrization

For every $t \geq 1$ and every nonnegative cost sequence $\mathbf{c}$,

$$CC_t^*(G, \mathbf{c}) \leq \max\{CC_t^*(G(U \rightarrow V), \mathbf{c}), CC_t^*(G(V \rightarrow U), \mathbf{c})\}, \quad (2.1)$$

and

$$CD_t^*(G, \mathbf{c}) \leq \max\{CD_t^*(G(U \rightarrow V), \mathbf{c}), CD_t^*(G(V \rightarrow U), \mathbf{c})\}. \quad (2.2)$$

2.1 The K_r -packing functional as a decomposition cost

Fix $r \geq 3$, set $t = 2$, and take

$$c_2 = c_r = 1, \quad c_i = \binom{i}{2} \quad (i \neq 2, r). \quad (2.3)$$

For $r = 3$, this reads

$$c_2 = c_3 = 1, \quad c_i = \binom{i}{2} \quad (i \geq 4),$$

so the triangle convention is exactly the case $r = 3$ of (2.3).

A clique of order $i \notin \{2, r\}$ can be replaced by its constituent edges at the same cost. Thus an optimal decomposition may be taken to use only edges and copies of K_r .

If the total weight assigned to K_r 's is y , then their total edge load is

$$\binom{r}{2} y.$$

The remaining edge weight has total

$$|E(G)| - \binom{r}{2} y,$$

and the decomposition cost is

$$|E(G)| - \left(\binom{r}{2} - 1 \right) y.$$

Maximizing y gives

$$CD_2^*(G, \mathbf{c}) = |E(G)| - \left(\binom{r}{2} - 1 \right) \nu_r^*(G) = \Phi_r^*(G). \quad (2.4)$$

2.2 Direct triangle specialization

For completeness, the case $r = 3$ also has a short direct proof.

Let u, v be nonadjacent vertices. Copying the incident weights of an optimal fractional triangle cover from u to a new false twin v , and then performing the reverse copying, gives two feasible covers whose costs sum to twice the original cost. Together with the corresponding edge-count identity,

$$\Phi_3^*(G_{v \rightarrow u}) + \Phi_3^*(G_{u \rightarrow v}) \geq 2\Phi_3^*(G). \quad (2.5)$$

This is included only to make the triangle application self-contained; the general argument uses Lemma 2.1.

3. Structural lemmas for constrained symmetrization

A vertex is simplicial when its open neighborhood is a clique.

Lemma 3.1

If one vertex of a clone class is simplicial, then every vertex in that class is simplicial and their common open neighborhood is a clique.

Proof

All vertices in the class have the same open neighborhood. $\square$

Lemma 3.2 — Chordality preservation

Let G be chordal. Let U, V be nonadjacent clone classes, and suppose that the vertices of V are simplicial. Then $G(U \rightarrow V)$ is chordal.

Proof

Delete U . The resulting graph is chordal. The common neighborhood of V is a clique. Add the vertices of U back one at a time with that same clique neighborhood. Adding a vertex with clique neighborhood preserves chordality. $\square$

Lemma 3.3 — Clone-class progress

Suppose U, V are distinct maximal clone classes with no edges between them. In either $G(U \rightarrow V)$ or $G(V \rightarrow U)$, the two classes merge and no old clone class is split. Hence the number of maximal clone classes strictly decreases.

Proof

After symmetrization, every vertex of $U \cup V$ has the same open neighborhood.

Let W be any clone class disjoint from $U \cup V$. Every two vertices of W have the same adjacency to the source class, and the operation changes their adjacency to the target class in the same way. Thus they remain clones. Other classes may merge, but none splits. $\square$

4. Complete-split terminal structure

Call a chordal graph **simplicially homogeneous** if every two nonadjacent simplicial vertices have the same open neighborhood.

Theorem 4.1 — Terminal characterization

A chordal graph is simplicially homogeneous if and only if it is complete split.

Proof

Every complete split graph is simplicially homogeneous: its only nonadjacent simplicial vertices lie in the independent set and all have the same neighborhood.

Conversely, suppose G is simplicially homogeneous.

If G is disconnected and one component contains an edge, choose a simplicial vertex with nonempty neighborhood in that component and a simplicial vertex in another component. They are nonadjacent and have different neighborhoods, a contradiction. Hence a disconnected simplicially homogeneous chordal graph is edgeless and is $S_{0,n}$.

Assume that G is connected. If G is complete, it is $S_{n,0}$. Otherwise, let $\mathcal{T}$ be a clique tree. It has at least two leaf maximal cliques.

For a leaf clique L , let L' be its unique neighboring clique and put

$$P_L = L \setminus L'.$$

The set P_L is nonempty. Every $x \in P_L$ occurs in no maximal clique other than L , so x is simplicial and

$$N_G(x) = L \setminus \{x\}. \quad (4.1)$$

We claim that P_L is a singleton. Choose a different leaf clique M and a vertex $y \in P_M$. If P_L contained distinct vertices x, x' , then both pairs x, y and x', y would be nonadjacent. Homogeneity would give

$$N_G(x) = N_G(y) = N_G(x'). \quad (4.2)$$

But $x, x' \in L$, so $x' \in N_G(x)$, while $x' \notin N_G(x')$, contradicting (4.2).

Write

$$P_L = \{x_L\}.$$

Then

$$N_G(x_L) = L \cap L'. \quad (4.3)$$

Private vertices from distinct leaf cliques are nonadjacent. Their neighborhoods are therefore equal. Hence every leaf separator in (4.3) is the same clique; call it K .

Every leaf bag contains K . For $a \in K$, the bags containing a form a connected subtree containing every leaf of $\mathcal{T}$. A connected subtree of a finite tree containing every leaf is the whole tree. Thus K lies in every maximal clique and is universal in G .

Put

$$J = G - K.$$

Every leaf-private vertex is isolated in J . If J contained an edge, a nontrivial connected component of the chordal graph J would contain a simplicial vertex z with nonempty neighborhood in J . Since K is universal,

$$N_G(z) = K \dot{\cup} N_J(z)$$

is a clique, so z is simplicial in G .

A leaf-private vertex x_L is nonadjacent to z , but

$$N_G(x_L) = K \neq K \dot{\cup} N_J(z) = N_G(z),$$

contradicting homogeneity. Therefore J is independent and

$$G = K \vee \bar{J}.$$

Thus G is complete split. $\square$

5. Complete-split reduction

Proof of Theorem 1.1

Start with a chordal graph G .

If G is complete split, stop. Otherwise, Theorem 4.1 and its contrapositive give non-adjacent simplicial vertices u, v with distinct open neighborhoods.

Let U, V be their maximal clone classes. They are nonadjacent and simplicial. Clone-symmetrization monotonicity gives a direction, either

$$G(U \rightarrow V) \quad \text{or} \quad G(V \rightarrow U),$$

whose F -value is at least $F(G)$.

The selected graph is chordal by Lemma 3.2 and has strictly fewer maximal clone classes by Lemma 3.3.

Repeat. The number of maximal clone classes decreases at every step, so the process terminates after at most $n - 1$ steps. The terminal graph is simplicially homogeneous and hence complete split by Theorem 4.1. $\square$

Corollaries 1.2 and 1.3 follow from Lemma 2.1 and identity (2.4).

Corollary 5.1 — Threshold sandwich

Let $\mathcal{T}_{\langle n}$ be the family of threshold graphs on the fixed n -vertex set. For every isomorphism-invariant clone-symmetrization-monotone graph parameter F ,

$$\boxed{\max_{G \in \mathcal{C}_n} F(G) = \max_{G \in \mathcal{T}_{\langle n}} F(G) = \max_{0 \leq p \leq n} F(S_{p, n-p}).} \quad (5.1)$$

Proof

A threshold graph can be constructed by repeatedly adding an isolated or a universal vertex. To construct $S_{p,q}$, first add q isolated vertices and then add p universal vertices. Hence

$$\mathcal{Q}_n \subseteq \mathcal{T}_{\langle n} \subseteq \mathcal{C}_n.$$

Theorem 1.1 gives

$$\max_{\mathcal{C}_n} F = \max_{\mathcal{Q}_n} F,$$

and the inclusions force equality throughout. $\square$

This applies in particular to CC_t^* , CD_t^* , and Φ_r^* .

Algorithm 5.2 — Fixed- r construction

Fix $r \geq 3$. Given a chordal graph G , repeat while the current graph is not complete split:

1. use the contrapositive of Theorem 4.1 to find nonadjacent simplicial vertices u, v with distinct neighborhoods;
2. let U, V be their maximal clone classes;
3. form $G(U \rightarrow V)$ and $G(V \rightarrow U)$;
4. compute Φ_r^* in both graphs and retain a direction of maximum value.

Proof of Corollary 1.7

There are at most $n - 1$ iterations.

For a graph J , $\nu_r^*(J)$ is the optimum of the explicit packing LP with one variable for each copy of K_r and one capacity constraint for each edge. For fixed r , it has at most

$$\binom{n}{r} = O(n^r)$$

variables and

$$\binom{n}{2} = O(n^2)$$

constraints.

The LP has rational 0-1 data and is solvable exactly in polynomial time. Clone classes, simplicial vertices, the two cloned graphs, and complete-split recognition are also computable in polynomial time. Therefore Algorithm 5.2 is polynomial for fixed r . $\square$

The scope restrictions are recorded in Section 12.2.

6. The complete-split orbit program

The structural part of the paper is now complete. All remaining quantitative statements reduce to evaluating the chosen parameter on

$$S_{p,q} = K_p \vee \overline{K}_q.$$

For the fixed-motif functional Φ_r^* , this evaluation is the following two-orbit linear program.

Fix $r \geq 3$ and write

$$S_{p,q} = K_p \vee \overline{K}_q.$$

Every K_r in $S_{p,q}$ has one of two types:

1. it is contained in K_p ;
2. it contains one independent vertex and $r - 1$ clique vertices.

Put

$$m = \binom{r}{2}, \quad c = \binom{r-1}{2}, \quad P = \binom{p}{2}. \quad (6.1)$$

After averaging over

$$\text{Sym}(p) \times \text{Sym}(q),$$

let α be the total packing weight on the first type and β the total packing weight on the second type.

For $p \geq r$, the exact orbit LP is

$$\max \alpha + \beta \tag{6.2}$$

subject to

$$m\alpha + c\beta \leq P, \tag{6.3}$$

$$(r-1)\beta \leq pq, \tag{6.4}$$

and

$$\alpha, \beta \geq 0. \tag{6.5}$$

Constraint (6.3) is the clique-edge capacity, while (6.4) is the cross-edge capacity. Proposition 6.1 proves directly, by orbit counting, that these two inequalities are exactly equivalent to all individual edge constraints after averaging.

The breakpoint has a direct geometric interpretation. A unit of the mixed variable β uses only

$$c = \binom{r-1}{2}$$

units of clique-edge capacity, compared with

$$m = \binom{r}{2}$$

for a unit of the internal variable α , but it also requires $r-1$ cross edges. Thus the LP uses mixed cliques as aggressively as the cross-edge reservoir permits. The equality

$$\frac{pq}{r-1} = \frac{P}{c} \iff (r-2)q = p-1 \tag{6.6}$$

is the exact point at which the available cross-edge capacity is just large enough for mixed K_r 's alone to exhaust the clique-edge capacity.

Above this threshold, the clique-edge constraint is saturated with β and the optimum has $\alpha = 0$. Below it, the cross-edge constraint binds first; the unused clique-edge capacity must then be filled by the residual internal variable $\alpha > 0$.

Proposition 6.1 — Exact fractional packing

For $p \geq r$,

$$\nu_r^*(S_{p,q}) = \frac{P}{m} + \left(\frac{P}{c} - \frac{P}{m} \right) \min \left\{ 1, \frac{(r-2)q}{p-1} \right\}. \tag{6.7}$$

If $p = r-1$ and $q \geq 1$, then

$$\nu_r^*(S_{p,q}) = 1. \tag{6.8}$$

If $p < r-1$, or if $p = r-1, q = 0$, then

$$\nu_r^*(S_{p,q}) = 0. \tag{6.9}$$

Proof

Assume first that $p \geq r$. The feasible region and objective of the full fractional packing LP are invariant under

$$\text{Sym}(p) \times \text{Sym}(q).$$

Averaging an optimal solution over this group therefore preserves feasibility and total weight and produces a symmetric optimum. Let α be the total weight on the $\binom{p}{r}$ internal copies of K_r , distributed uniformly over that orbit, and let β be the total weight on the

$$q \binom{p}{r-1}$$

mixed copies, also distributed uniformly.

Every clique edge lies in $\binom{p-2}{r-2}$ internal copies. Its load from the internal orbit is therefore

$$\frac{\alpha \binom{p-2}{r-2}}{\binom{p}{r}} = \frac{\binom{r}{2}}{\binom{p}{2}} \alpha = \frac{m\alpha}{P}.$$

The same clique edge lies in

$$q \binom{p-2}{r-3}$$

mixed copies, so its mixed load is

$$\frac{\beta q \binom{p-2}{r-3}}{q \binom{p}{r-1}} = \frac{\binom{r-1}{2}}{\binom{p}{2}} \beta = \frac{c\beta}{P}.$$

Thus every clique-edge capacity constraint is exactly

$$m\alpha + c\beta \leq P.$$

Every cross edge lies in $\binom{p-1}{r-2}$ mixed copies. Its load is

$$\frac{\beta \binom{p-1}{r-2}}{q \binom{p}{r-1}} = \frac{r-1}{pq} \beta,$$

so every cross-edge capacity constraint is exactly

$$(r-1)\beta \leq pq.$$

Conversely, if $\alpha, \beta \geq 0$ satisfy these two inequalities, the uniform assignments above give a feasible fractional packing. Hence the full packing LP is exactly equivalent to (6.2)–(6.5).

Since

$$c = \binom{r-1}{2} < \binom{r}{2} = m,$$

a unit of β uses less clique-edge capacity than a unit of α . An optimum therefore maximizes β first:

$$\beta = \min \left\{ \frac{P}{c}, \frac{pq}{r-1} \right\},$$

and fills the remaining clique-edge capacity with

$$\alpha = \frac{P - c\beta}{m}.$$

The resulting objective value is (6.7).

If $p = r - 1$ and $q \geq 1$, every K_r consists of all clique vertices and one independent vertex. Any two such copies share every clique edge, so the fractional packing number is 1. If $p < r - 1$, or if $p = r - 1, q = 0$, no copy of K_r exists. $\square$

Corollary 6.2 — Exact form of Φ_r^*

For $p \geq r$, there are two regimes.

If

$$(r - 2)q \geq p - 1, \quad (6.10)$$

then

$$\Phi_r^*(S_{p,q}) = pq - \frac{p(p-1)}{r-1}. \quad (6.11)$$

If

$$(r - 2)q \leq p - 1, \quad (6.12)$$

then

$$\Phi_r^*(S_{p,q}) = \frac{\binom{p}{2} + pq}{\binom{r}{2}}. \quad (6.13)$$

The two expressions agree on the boundary. Formula (6.11) also remains valid for $p = r - 1, q \geq 1$. For $p < r - 1$, $\Phi_r^*(S_{p,q})$ is simply the edge count.

7. Exact triangle maximum

Put $r = 3, q = n - p$, and

$$\Phi^*(G) = |E(G)| - 2\nu_3^*(G).$$

In the mixed-saturated regime $q \geq p - 1$,

$$\Phi^*(S_{p,q}) = pq - \binom{p}{2}. \quad (7.1)$$

Completing the square gives

$$pq - \binom{p}{2} = \frac{(2n+1)^2 - (6p - (2n+1))^2}{24}. \quad (7.2)$$

The minimum possible value of

$$|6p - (2n+1)|$$

is 1 when $3 \nmid 2n+1$, and 3 when $3 \mid 2n+1$. Since an odd square is congruent to 1 or 9 (mod 24), the maximum of (7.2) over integers p is

$$\left\lfloor \frac{(2n+1)^2}{24} \right\rfloor. \quad (7.3)$$

The maximizing integer lies near $(2n+1)/6$ and satisfies $q \geq p - 1$.

In the unsaturated regime $q \leq p - 1$,

$$\Phi^*(S_{p,q}) = \frac{\binom{p}{2} + pq}{3} = \frac{p(2n - p - 1)}{6}. \quad (7.4)$$

For $n \geq 3$, this is at most $n(n-1)/6$, which is strictly below the value in (7.3). The cases $n = 1, 2$ are immediate.

This proves Theorem 1.4.

In particular,

$$\max_{G \in \mathcal{C}_n} \Phi^*(G) = \frac{n^2}{6} + \frac{n}{6} + O(1). \quad (7.5)$$

8. Exact finite optimization for K_r

Fix $r \geq 3$, put $q = n - p$, and abbreviate

$$\Psi_{r,n}(p) = \Phi_r^*(S_{p,n-p}).$$

Corollary 6.2 and the exceptional cases of Proposition 6.1 divide the optimization into three formal regimes. We show that only two can be globally extremal.

8.1 The K_r -free branch

For $p \leq r - 2$, the graph $S_{p,n-p}$ contains no K_r . Hence

$$\Psi_{r,n}(p) = \binom{p}{2} + p(n-p) = \frac{p(2n-p-1)}{2}. \quad (8.1)$$

This is increasing on the permitted integer interval. With

$$\ell = \min\{n, r-2\},$$

the branch maximum is therefore

$$L_{r,n} = \frac{\ell(2n-\ell-1)}{2}. \quad (8.2)$$

If $n < r$, this is $\binom{n}{2}$.

8.2 The saturated branch

In the saturated regime,

$$(r-2)(n-p) \geq p-1, \quad (8.3)$$

we have

$$A_{r,n}(p) = \Psi_{r,n}(p) = pn - \frac{r}{r-1}p^2 + \frac{p}{r-1}. \quad (8.4)$$

Completing the square gives

$$A_{r,n}(p) = \frac{((r-1)n+1)^2}{4r(r-1)} - \frac{r}{r-1}(p-p_0)^2, \quad (8.5)$$

where

$$p_0 = \frac{(r-1)n+1}{2r}. \quad (8.6)$$

For $n \geq r$, define

$$d_{r,n} = \min_{\substack{p \in \mathbb{Z} \\ p \geq r-1}} |p - p_0|. \quad (8.7)$$

The nearest constrained integer lies in the saturated interval (8.3). To see this, if $n < 2r$, then $p_0 < r-1$, so the constrained minimizer is $p = r-1$, which satisfies (8.3). If $n \geq 2r$, then $p_0 \geq r-1$ and the saturated upper endpoint is

$$p_{\text{sat}} = \frac{(r-2)n+1}{r-1}.$$

Indeed,

$$p_{\text{sat}} - p_0 - 1 = \frac{n(r^2 - 2r - 1) - 2r^2 + 3r + 1}{2r(r-1)} \geq 0$$

for $n \geq 2r$ and $r \geq 3$. Hence both nearest integers to p_0 are saturated. Therefore the exact saturated maximum is

$$A_{r,n}^{\#} = \frac{((r-1)n+1)^2}{4r(r-1)} - \frac{r}{r-1}d_{r,n}^2. \quad (8.8)$$

Equivalently, evaluate (8.4) at the one or two integers nearest to $\max\{r-1, p_0\}$.

8.3 The unsaturated branch is never a third extremizer

In the unsaturated regime,

$$(r-2)(n-p) \leq p-1,$$

Corollary 6.2 gives

$$B_{r,n}(p) = \frac{p(2n-p-1)}{r(r-1)}. \quad (8.9)$$

This is nondecreasing for integers $p \leq n$. Hence every unsaturated value is at most

$$U_{r,n} = \frac{n(n-1)}{r(r-1)}. \quad (8.10)$$

Suppose first that $r \leq n \leq 2r-1$. The K_r -free value at $p = r-2$ is

$$L_{r,n} = \frac{(r-2)(2n-r+1)}{2}.$$

After clearing denominators,

$$2r(r-1)(L_{r,n} - U_{r,n}) = r(r-1)(r-2)(2n-r+1) - 2n(n-1). \quad (8.11)$$

The right-hand side is a concave quadratic in n . At $n = r$ it equals

$$r(r-1)(r^2 - r - 4) > 0,$$

and at $n = 2r-1$ it equals

$$(r-1)(r(r-2)(3r-1) - 4(2r-1)) > 0.$$

Thus it is nonnegative throughout the interval, and

$$U_{r,n} \leq L_{r,n}. \quad (8.12)$$

Now suppose $n \geq 2r$. Then $p_0 \geq r-1$. Rounding the concave quadratic (8.5) loses at most $r/[4(r-1)]$, so

$$A_{r,n}^{\#} \geq \frac{((r-1)n+1)^2}{4r(r-1)} - \frac{r}{4(r-1)}.$$

Subtracting (8.10) gives

$$A_{r,n}^{\#} - U_{r,n} \geq \frac{(r-3)(r+1)n^2 + 2(r+1)n + 1 - r^2}{4r(r-1)}. \quad (8.13)$$

For $r = 3$, the numerator is $8(n-1) > 0$; for $r \geq 4$, it is plainly positive. Hence

$$U_{r,n} \leq A_{r,n}^{\#}. \quad (8.14)$$

The unsaturated branch never creates a third global maximum.

Combining Sections 8.1–8.3 proves Theorem 1.5:

$$\max_{G \in \mathcal{C}_n} \Phi_r^*(G) = \begin{cases} L_{r,n}, & n < r, \\ \max\{L_{r,n}, A_{r,n}^\#\}, & n \geq r. \end{cases} \quad (8.15)$$

For fixed r , the K_r -free candidate is $O_r(n)$, while

$$A_{r,n}^\# = \frac{r-1}{4r}n^2 + O_r(n).$$

Thus

$$\max_{G \in \mathcal{C}_n} \Phi_r^*(G) = \frac{r-1}{4r}n^2 + O_r(n). \quad (8.16)$$

For $r = 3$, (8.15) reduces to the closed form in Theorem 1.4.

9. Integral application

Let

$$M_r = \binom{r}{2}.$$

Every edge partition into copies of K_r and single edges corresponds to an edge-disjoint K_r -packing, and therefore

$$\pi_r(G) = |E(G)| - (M_r - 1)\nu_r(G). \quad (9.1)$$

For fixed r , the Haxell–Rödl theorem, or Yuster’s family version, gives uniformly over n -vertex graphs

$$\nu_r^*(G) - \nu_r(G) = o(n^2). \quad (9.2)$$

Hence

$$\pi_r(G) \leq \Phi_r^*(G) + o(n^2).$$

The asymptotic consequence (8.16) of Theorem 1.5 gives the corresponding upper bound over chordal graphs.

Conversely,

$$\pi_r(G) \geq \Phi_r^*(G)$$

for every graph. Applying this to a complete split graph maximizing Φ_r^* gives the matching lower bound. This proves Corollary 1.6.

For $r = 3$, any partition into triangles and edges is a clique partition, so

$$\text{cp}(G) \leq \pi_3(G).$$

Together with the classical complete split lower-bound construction, this settles the Erdős–Ordman–Zalcstein asymptotic constant.

10. Fractional scope and integral obstruction

The monotonicity relies on convex averaging and has no integral analogue.

Consider two triangles

$$abc, \quad cde$$

sharing only the vertex c . The vertices a, d are nonadjacent and simplicial, and

$$\nu_3(G) = 2.$$

After cloning d to a , or a to d , the resulting triangles share an edge, so both integral packing numbers are 1.

Thus the exact reduction must occur at the fractional level. The $o(n^2)$ term enters only through (9.2).

11. Supplementary computational regression tests

Every statement tested in this section is proved analytically in the manuscript. The supplied scripts are optional regression tests and reproducibility material; no computation, LP solver, finite graph atlas, or finite enumeration is a logical premise of any theorem. They provide the following independent checks:

1. all chordal graph-atlas graphs through seven vertices symmetrize to complete split graphs;
2. both clone directions preserve chordality;
3. maximal clone classes strictly decrease;
4. Φ_r^* never decreases for $r = 3, 4$;
5. chordal and complete-split maxima agree through seven vertices;
6. Proposition 6.1 agrees with the full packing LP for $3 \leq r \leq 6$ and $0 \leq p, q \leq 8$;
7. Theorem 1.4 holds for $1 \leq n \leq 500$;
8. the exact finite formula in Theorem 1.5 agrees with direct maximization over all $0 \leq p \leq n$ for $3 \leq r \leq 30$ and $1 \leq n \leq 500$;
9. the fixed- r leading coefficient agrees numerically for $3 \leq r \leq 8$.

The corresponding analytic proofs are Theorem 4.1, Lemmas 2.1 and 3.1–3.3, Proposition 6.1, Sections 7–8, and the standard external theorems cited in Section 2. The computations are supplementary evidence only.

12. Discussion

12.1 Contributions

The fractional clone-symmetrization mechanism is due to Balogh et al. The contribution of this paper is the constrained chordal analysis:

1. simplicial clone-class symmetrization remains inside the chordal class;
2. maximal clone classes give a strictly decreasing finite potential;
3. the terminal chordal graphs are exactly complete split;
4. every fractional parameter in Corollary 1.2 reduces to the $n + 1$ graphs $S_{p, n-p}$, and therefore also to the threshold-graph class;
5. fixed clique motifs admit an explicit two-orbit linear program;

6. that program yields the exact finite K_r maximum, its triangle specialization, the fixed- r asymptotics, and the fixed- r constructive reduction.

An earlier companion manuscript developed a direct fractional analysis of split graphs. The complete-split reduction and exact triangle theorem now make that manuscript unnecessary as a logical dependency of the present result, although the direct method may retain independent methodological interest.

12.2 Scope and limitations

The claims of the paper are intentionally limited as follows.

- The general fractional symmetrization lemma is attributed to Balogh et al.; it is not claimed here as a new method.
- The claimed structural contribution is the constrained chordal process and its complete-split terminal family. A targeted search found no earlier reduction of this form, but forward-citation tracking and specialist review remain necessary.
- The algorithmic statement is polynomial only for fixed r . It is not polynomial jointly in (G, r) , and no comparable claim is made for arbitrary unrestricted cost sequences in Corollary 1.2.
- The algorithm constructs a fractional complete-split extremizer; it does not construct the final integral partition supplied asymptotically by Haxell–Rödl/Yuster.
- The exact extremizer theorem does not imply stability or characterize all near-extremizers.
- Apart from the cited mathematical theorems, the complete proof is analytic. No script, solver output, or finite enumeration is used as a premise.
- The supplementary computational regressions are independent evidence and are not proof dependencies.

12.3 Future directions

The one-parameter reduction suggests several separate questions.

1. **Simonovits-type stability.** Determine whether, for every fixed $r \geq 3$ and every $\varepsilon > 0$, there is a $\delta = \delta(r, \varepsilon) > 0$ such that every n -vertex chordal graph satisfying

$$\Phi_r^*(G) \geq \max_{H \in \mathcal{C}_n} \Phi_r^*(H) - \delta n^2$$

is within εn^2 edge edits of some maximizing complete split graph $S_{p, n-p}$, or converges to that family in an equivalent graphon metric.

A proof would have to quantify the gain under nonterminal symmetrizations and exclude large structural changes with negligible objective improvement.

2. Determine exact integer formulas for π_r .
3. Develop implementations that avoid repeated general LP solves.
4. Identify other hereditary graph classes whose constrained symmetrization has a canonical terminal family.

These questions are independent of the present proof.

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References

1. J. Balogh, J. He, R. A. Krueger, T. Nguyen, and M. C. Wigal, *Clique covers and decompositions of cliques of graphs*, arXiv:2412.05522v2, 2024.
2. E. A. Bender, L. B. Richmond, and N. C. Wormald, *Almost all chordal graphs split*, Journal of the Australian Mathematical Society (Series A) **38** (1985), 214–221. DOI: [link](#).
3. G.-T. Chen, P. Erdős, and E. T. Ordman, *Clique Partitions of Split Graphs*, in **Combinatorics, Graph Theory, Algorithms and Applications** (Beijing, 1993), World Scientific, 1994, 21–30.
4. V. Chvátal and P. L. Hammer, *Aggregation of inequalities in integer programming*, in **Studies in Integer Programming**, Annals of Discrete Mathematics **1**, North-Holland, 1977, 145–162.
5. G. A. Dirac, *On rigid circuit graphs*, Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg **25** (1961), 71–76. DOI: [link](#).
6. P. Erdős, E. T. Ordman, and Y. Zalcstein, *Clique Partitions of Chordal Graphs*, Combinatorics, Probability and Computing **2** (1993), 409–415. DOI: [link](#).
7. P. E. Haxell and V. Rödl, *Integer and Fractional Packings in Dense Graphs*, Combinatorica **21** (2001), 13–38. DOI: [link](#).
8. A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986.
9. R. Yuster, *Integer and Fractional Packing of Families of Graphs*, Random Structures & Algorithms **26** (2005), 110–118. DOI: [link](#).
10. A. A. Zykov, *On some properties of linear complexes*, Matematicheskii Sbornik (N.S.) **24(66)** (1949), 163–188.